CONNECTAFLOW

Platform Architecture & Feature Specification

Complete Engineering & Product Design Document

Version 2.0 | GTM Intelligence → List Enrichment → Messaging → Execution → Analytics

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Every feature decision is made at the intersection of two questions: Does a senior sales strategist recognise this as the right way to think about outbound? And does an engineer recognise this as clean, scalable, and maintainable? Both must be true. 8

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The AI drafts everything. The human edits and approves. The output standard matches the SalesHive playbook format: persona-specific copy, full sequence with 3–5 variants per testable variable, and a clean A/Z test configuration ready for Smartlead. 23

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For each sequence step, the system generates the full assembled email PLUS each named component broken out individually for editing and A/Z variant

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Campaign Assembly is where a named, structured GTM hypothesis is packaged and launched. It connects a list segment to a messaging set and sequence, configures the A/Z test, validates volume, and pushes the complete package to Smartlead via API. Every campaign that leaves ConnectaFlow is a well-defined, traceable experiment. 27

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The Analytics module pulls campaign performance data from Smartlead and enriches it with ConnectaFlow's structural metadata — ICP, signal segment, messaging variant, play type. The result is not just a metrics dashboard but a strategic learning layer: you can see not just your open rate but which ICP, which signal, which messaging angle, and which A/Z variant drove it. 29

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The quality of AI-generated ICPs, personas, and email copy is directly proportional to the richness of the Section A input. The system should prompt users to be specific — vague inputs produce generic outputs. Consider a "context quality score" that tells users how complete their GTM Intelligence setup is before they proceed. 35

Enrichment Pipeline Architecture 35

Build enrichment as an async queue. Large lists (1,000+ companies) should not block the UI. Show real-time progress per company. Allow partial enrichment — the user should be able to work with the first 100 scored companies while the rest are still processing. 35

Smartlead Sync 35

ConnectaFlow passes metadata tags (ICP_ID, SEGMENT_ID, MESSAGING_ID, VARIANT_ID) to Smartlead campaigns and contacts. When

pulling analytics, use these tags to join Smartlead data back to ConnectaFlow objects. This join is the engine of the analytics layer — design it deliberately.35

A/Z Test Integrity 35

Enforce single-variable testing at the data model level, not just the UI. A campaign record has one AZ_TEST_ID with one test variable. Multiple variable changes in the same campaign must be blocked at the API layer, not just warned about in the frontend. 35

Custom AI Columns — Rate Limiting 35

Custom AI column runs against large lists can be expensive and slow. Implement cost estimation before run, rate limiting per workspace, and incremental processing (run on High fit companies first). Cache company-level results so the same company is not re-enriched unnecessarily across multiple lists. 35

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Built at the intersection of world-class B2B sales practice and engineering excellence. 35

00 Platform Philosophy

What ConnectaFlow is and what it is not

What This Platform Is

ConnectaFlow is an outbound intelligence operating system — built by engineers who understand software at scale, designed by people who run world-class B2B sales campaigns every day. It is the infrastructure layer that connects GTM strategy, list intelligence, messaging, execution, and analytics into a single coherent workflow.

Every feature decision is made at the intersection of two questions: Does a senior sales strategist recognise this as the right way to think about outbound? And does an engineer recognise this as clean, scalable, and maintainable? Both must be true.

Core Design Principles

Speed	The platform should feel faster than doing it manually at every step. AI generates, humans approve. Nothing feels slow.
Discipline	Every campaign is a named hypothesis. Volume minimums are enforced. One variable is tested at a time. The system creates GTM rigour without being a blocker.
Intelligence	Everything is connected by IDs. Any result can be traced back to ICP, signal, messaging variant, and play type. Analytics are meaningful because the data model is structured from day one.
Separation	ConnectaFlow owns intelligence, enrichment, messaging, and analytics. Smartlead owns deliverability and execution. These lines never blur.
Flexibility	The platform can be used in one sitting or over multiple sessions. No enforced sequence. No artificial waiting. Use it as fast or as deliberately as the work demands.

What ConnectaFlow Is NOT

- A real-time signal-to-send automation tool — signals inform list curation, not send triggers
- A list building tool — Apollo, Clay, and similar tools remain external; ConnectaFlow enriches and scores
- A CRM — contact management and deal tracking are out of scope
- A Smartlead replacement — execution, deliverability, and inbox management stay in Smartlead
- A multi-channel sequencer — email is the primary channel at MVP; LinkedIn is V2

01 System Architecture

How the platform is structured end to end

High-Level Module Map

01	**GTM Intelligence** Client context → ICPs, personas, triggers, signals, motion intent
02	**List Ingestion** Upload CSV from Apollo/Clay → column mapping → deduplication
03	**Enrichment Engine** AI columns, signal detection, ICP scoring, company profiling
04	**Messaging Studio** AI-generated sequences, variable architecture, A/Z config, versioning
05	**Campaign Assembly** Segment selection, hypothesis writing, volume check, Smartlead push

06

Analytics

Smartlead metrics pull, ICP/signal/variant slicing, campaign review

Data Flow

The following IDs connect every object in the system. No piece of data is ever orphaned — everything traces back to its origin and forward to its outcome.

ICP_ID	ICP Definition	Personas, Triggers, Signals, Lists, Campaigns, Analytics
PERSONA_ID	Buyer Persona	ICP, Messaging Sets, Campaigns
TRIGGER_ID	Buying Trigger	ICP, Signal Definitions
SIGNAL_ID	Detectable Signal	Trigger, Enrichment output, Company tags, Campaign segmentation
LIST_ID	Uploaded Company List	ICP, Companies, Enrichment runs
COMPANY_ID	Individual Company (hashed domain)	List, ICP Score, Signal tags, Enrichment columns
SEGMENT_ID	Filtered subset of a list	List, ICP score band, Signal tags, Campaigns
MESSAGING_ID	Approved Messaging Set	ICP, Persona, Motion Intent, Campaigns
SEQUENCE_ID	Step structure + timing config	Messaging Set, Campaign
VARIANT_ID	Individual A/Z variant	Sequence step, Variable tested, Campaign
PLAY_ID	Play / Motion type	ICP, Campaign
CAMPAIGN_ID	Assembled campaign package	ICP, Segment, Messaging, Sequence, Hypothesis, Smartlead ID
AZ_TEST_ID	A/Z test config for a campaign	Campaign, Variant, Results
SMARTLEAD_ID	Smartlead campaign reference	Campaign, Analytics pull

02 GTM Intelligence Module

Client context → structured targeting strategy

Purpose

The GTM Intelligence module translates raw client context — product information, market knowledge, customer examples, competitive landscape — into a fully structured targeting strategy. It outputs ICPs, personas, buying triggers, signal definitions, and motion intent that drive every downstream module. Nothing in the platform works well without a well-built GTM Intelligence foundation.

This module is AI-assisted and human-approved. The AI generates suggestions from the inputs. The user edits, rejects, or adds to every output. Nothing is locked — every field is editable at any time.

Section A — Company Context Input

The foundational input layer. Supports both structured fields and free-form file upload. The richer this input, the higher quality all downstream outputs.

Company Name	Text	Used to pre-populate all generated copy
Website URL	Text / URL	Used for AI website analysis in enrichment
Product Description	Long text	Plain language — what it does, not feature list
Core Problem Solved	Long text	The pain, not the product
Product Category	Dropdown + freetext	e.g. Sales Engagement, HR Tech, RevOps
Pricing Model	Dropdown	Per seat / usage-based / flat / custom enterprise
Typical Deal Size	Range input	Shapes ICP employee/revenue targeting downstream

Best Current Customers	Repeating field (3–5)	Company name + why they bought
Competitors / Alternatives	Repeating field	What prospects use instead — critical for competitive plays
Geographic Focus	Multi-select	Regions and countries for ICP and list filtering
Additional Context Files	File upload (multi)	Pitch deck, one-pager, battlecard, case studies, previous campaign results — PDF, DOCX, PPTX

File uploads are parsed by AI and their content is used to enrich all generated outputs — the system reads pitch decks, extracts value props, and incorporates case study evidence into generated persona pain maps and email copy.

Section B — ICP Builder

AI generates 3–5 ICP suggestions based on Section A. Each ICP is a single structured sentence following the agency-defined format. The user accepts, edits, deletes, or adds ICPs freely.

ICP Sentence Structure

Geography	Target region or country	US, UK, DACH, ANZ
Industry	Primary industry segment	SaaS, Professional Services, FinTech
Company Condition	Size, stage, or model	50–200 employees, Series A/B, bootstrapped
Trigger Context	Situation creating potential need	hiring SDRs, recently funded, new CRO hired

Example ICP	US SaaS companies with 50–200 employees actively hiring SDRs

ICP Card Fields

- ICP Name — short label for internal reference (e.g. "Outbound-Stage SaaS")
- ICP Statement — the structured sentence above

- Firmographic Range — employee count, revenue range, business model, geography
- Priority — Primary / Secondary / Experimental
- Rationale — AI-generated explanation of why this ICP fits the product (editable)
- List Sourcing Guidance — auto-generated Apollo/Clay filter instructions per ICP

Section C — Persona Definitions

Per ICP, the user defines 1–2 target personas. AI generates suggestions pulled from company context. Each persona card maps directly to the SalesHive buyer group format — titles, pains, KPIs, and decision role.

Persona Name	e.g. "VP Sales", "CRO", "Head of RevOps"
Department	e.g. Sales, Marketing, Revenue Operations
Seniority	C-Suite / VP / Director / Manager / Individual Contributor
Common Titles	Full list used for contact sourcing in Apollo/Clay (downstream)
Responsibilities	What they own day-to-day
KPIs	What they are measured on — directly informs messaging angle
Pain Points (3–5)	Specific pains mapped to the client's product value prop
How We Solve Each Pain	Per pain, one sentence of product-specific resolution
Insight to Provide	A non-obvious truth about their situation that frames your product
Thoughtful Opening Question	One question that opens a conversation without feeling like a pitch
Decision Role	Champion / Decision Maker / Influencer / Blocker
Common Objections	Pre-loaded from agency playbook, mapped to suggested responses

Section D — Buying Triggers + Signal Mapping

Triggers are business situations that create urgency or need. Each trigger is mapped to an observable signal that can be detected in the Enrichment module. This linkage is the foundation of signal-informed list segmentation.

Key Principle	Signals do NOT trigger sends. Signals inform list quality and messaging angle. A company tagged with recently_funded gets prioritised in the segment and receives signal-aware copy — it does not get auto-emailed.

Trigger Card Fields

- Trigger Name — short label
- Description — what is happening at the company
- Why It Matters — how this creates buying intent for the client's product
- Observable Signal → — the data point that indicates this trigger is firing
- Signal Type — Job posting / Funding event / Tech stack / Leadership hire / Website signal / News
- Signal Source — LinkedIn / Crunchbase / BuiltWith / Website scrape / Imported column
- Detection Method — Automated (API/scrape) / Manual (CSV column) / AI analysis

Section E — Motion Intent

Strategic intent per ICP. Not copy — that is written in the Messaging Studio. Motion Intent defines how you want to approach this ICP, informing tone, angle, and sequence structure. It is the strategic brief that the Messaging Studio reads.

Motion Name	Freetext	e.g. "Outbound Scaling Play", "New CRO Play"
Motion Type	Cold / Trigger-informed / Competitive displacement / Re-engagement	Determines sequence template used in Messaging Studio
Primary Angle	Pain-led / Insight-led / Social proof / Signal-led / Competitive	Drives the Opener and Problem variables in email copy
Target Persona	Linked PERSONA_ID	Which persona leads this motion
Tone	Direct / Consultative / Challenger / Conversational	Applied globally across sequence steps
CTA Intent	Meeting ask / Resource share / Question-based	Drives the CTA variable generation
Messaging Notes	Freetext	Angles to use, objections to pre-empt, things to avoid, social proof to include

GTM Intelligence Outputs

Output 1 — GTM Strategy Brief

An exportable, shareable document (PDF + shareable link) containing all ICPs, personas, triggers, signals, and motion intents. Used for internal alignment, client review, and onboarding new team members. Auto-generated. Regenerates when any component is updated.

Output 2 — List Sourcing Guide (per ICP)

Since list building is external to the platform, each ICP generates a concrete, copyable sourcing guide for Apollo or Clay:

Example	ICP-01: US SaaS, 50–200 employees, hiring SDRs | Apollo filters: Geography=US, Industry=Software/SaaS, Employees=50-200, Job keywords=SDR, Sales Development | Target titles: VP Sales, Head of Sales, CRO | Estimated list size: 500–2,000 companies

Play Type Catalogue

Pre-built play types informed by agency best practice. Users select from this catalogue in Campaign Assembly. Each play type carries default motion settings and sequence structure recommendations.

Cold Outreach	Standard account-based cold outreach. Pain-led or insight-led. Default 5-step sequence.
Trigger — Funded	Target recently funded companies. Signal-led opener referencing the funding event. Urgency-aware copy.
Trigger — Hired CXO	New VP/CRO/CMO in last 90 days. Leadership change angle. New exec establishing priorities.
Trigger — Hiring SDRs	Company actively hiring sales development reps. Sales scaling angle. High intent signal.
Competitive Displacement	Target competitor's known customers. Pain-led with competitive contrast. Requires competitor mapping in GTM Intelligence.
Tech Stack — Uses X	Companies using a specific tool (e.g. Salesforce, HubSpot). Integration or migration angle.
Local / Event	Segment by geography or event proximity. Personalised local reference in opener.
Champion Re-engagement	Prospects who previously engaged but went cold. Reference prior touch. Softer, warmer tone.
Referral / Partner	Outreach from a shared network or referral program. Warmest cold email — mutual connection referenced.

Content / Research	Lead with insight, survey, or piece of content. Lowest friction entry. Builds list for follow-up.
Inbound Follow-up	Prospects who filled a form, visited pricing page, or engaged with content but did not convert.
Founder-to-Founder	Outreach from founder/executive to founder/executive. Peer-to-peer tone. High-trust positioning.

03 List Ingestion Module

Upload external lists → map columns → validate → tag to ICP

Purpose

List building is external to ConnectaFlow — the platform does not source contacts. Users build lists in Apollo, Clay, LinkedIn Sales Navigator, or any other tool and upload them as CSVs. The List Ingestion module receives those uploads, maps columns, validates data quality, deduplicates, and tags each list to an ICP before passing to enrichment.

Upload Flow

1. User selects which ICP this list belongs to (required before upload)
2. User uploads CSV — no format restrictions, any column structure accepted
3. Column mapping UI — user maps their column headers to ConnectaFlow's canonical fields
4. Validation run — system flags missing required fields, malformed data, and duplicates
5. User reviews validation report — can fix, accept warnings, or re-upload
6. List is ingested, tagged with LIST_ID and ICP_ID, passed to Enrichment queue

Required vs Optional Fields

Company Name	✅ Required	Used for deduplication and enrichment lookup
Domain	✅ Required	Primary COMPANY_ID key — all enrichment anchored to domain
Employee Count	✅ Required	Core structured fit scoring field
Industry	✅ Required	Core structured fit scoring field
Geography (HQ Country)	✅ Required	Core structured fit scoring field
Revenue Range	Optional	Improves scoring accuracy when available
LinkedIn Company URL	Optional	Used for hiring signal detection
Tech Stack	Optional	Pre-imported tech data from Clay or BuiltWith
Funding Stage / Date	Optional	Pre-imported funding data reduces enrichment API calls
Contact Name	Optional	Contact-level data — stored but not required at company level
Contact Email	Optional	Passed to Smartlead at campaign push
Contact Title	Optional	Used for persona matching and segmentation
Custom columns (any)	Optional	Any additional columns are preserved and available as enrichment inputs or filter criteria

Deduplication Logic

- Primary key: domain (normalised — strips www, http, trailing slashes)
- Secondary check: company name fuzzy match for companies without domains
- Cross-list deduplication: a company already in the workspace from a prior list is flagged, not blocked
- Contact-level dedup: same email address across multiple contacts flagged
- User controls dedup behaviour — merge / skip / overwrite

04 Enrichment Engine

ICP scoring · signal detection · AI columns · company profiling

Purpose

The Enrichment Engine is the analytical core of ConnectaFlow. It takes a raw uploaded list and transforms it into a scored, tagged, profiled dataset that is ready for intelligent segmentation. It runs

automatically after list ingestion but can also be triggered manually, run incrementally, or refreshed on demand.

The engine has three parallel tracks: ICP Scoring, Signal Detection, and AI Column Generation. All three run concurrently and write results to the company record.

Track 1 — ICP Scoring

Score Formula

ICP Score	Final Score (0–100) = Structured Fit Score × 70% + Contextual Fit Score × 30%

Structured Fit Factors (70%)

These are deterministic, rule-based scores calculated from firmographic data:

Industry Match	Exact match = 100, parent category match = 70, adjacent = 40, no match = 0	30%
Employee Range Match	Within ICP range = 100, within 20% of range = 70, outside = scaled decay	25%
Geographic Match	Exact match = 100, regional match = 60, no match = 0	25%
Revenue Range	Within range = 100, estimated range acceptable, missing = neutral (50)	10%
Business Model Match	B2B/B2C/marketplace match against ICP definition	10%

Contextual Fit Factors (30%)

These are AI-assessed scores based on company description, website content, and product category analysis:

- Product category alignment — does what they sell match who you target?
- Customer segment alignment — do they sell to your personas' peer companies?
- Growth stage relevance — are they at the stage where they'd need your product?
- Sales motion relevance — do they run the type of sales motion your product serves?

Score Outputs Per Company

- Structured Fit Score (0–100)
- Contextual Fit Score (0–100)
- Final ICP Score (0–100)
- ICP Fit Category — High (75–100) / Medium (50–74) / Low (0–49) — thresholds configurable
- ICP Reasoning — 2–3 sentence AI explanation of the score (editable)

Track 2 — Signal Detection

Signals are detected per company based on the signal definitions created in GTM Intelligence. Each detected signal is stored as a tag on the company record with metadata about when it was detected and what source confirmed it.

Signal Tags (Applied Per Company)

recently_funded	Automated	Crunchbase / Dealroom API	Date of announcement
hiring_sdrs	Automated	LinkedIn Job Postings API	Active posting date
hiring_aes	Automated	LinkedIn Job Postings API	Active posting date
new_cxo_hired	Automated	LinkedIn Profile changes	Date of role change
headcount_growth_20pct	Automated	LinkedIn employee count delta	Rolling 12 months
tech_stack_match	Automated	BuiltWith / Wappalyzer	Last crawl date
competitor_customer	Manual / Imported	CSV column or G2/ Capterra scrape	Import date
website_signal	AI analysis	Company website crawl	Last crawl date
expansion_news	Automated	News API / press mentions	Article date
custom_signal	User-defined	Any configured source	Detection date

Track 3 — AI Column Generation

This is the most powerful and differentiating feature of the Enrichment Engine. Users can add custom AI-powered columns to their list — each column runs an AI analysis against available company data (website, description, LinkedIn, news) and writes a structured result per company.

This is modelled on Clay's AI column concept but purpose-built for outbound intelligence — every column is designed to feed downstream scoring, segmentation, or messaging personalisation.

Built-in AI Columns

Company Summary	Text (2–3 sentences)	What the company does in plain language — used in messaging personalisation	Website + LinkedIn
Business Model	Classification	B2B / B2C / B2B2C / Marketplace / SaaS / Services	Website AI analysis
Product Category	Classification	e.g. Sales Engagement, HR Tech, FinTech — aligns to ICP definition	Website + LinkedIn
Target Customer Type	Classification	Who they sell to — SMB / Mid-Market / Enterprise / Consumer	Website + LinkedIn
Growth Stage Estimate	Classification	Seed / Early / Growth / Scale / Mature — derived from headcount + funding	Multi-source
Sales Motion Type	Classification	Inbound-led / Outbound-led / PLG / Channel — informs contextual fit scoring	Website + job postings
Tech Stack Summary	Text list	Key tools the company uses — CRM, SEP, marketing automation	BuiltWith / Wappalyzer
Hiring Velocity	Score + trend	Rate of headcount growth over 6/12 months	LinkedIn delta
Recent News Summary	Text (1–2 sentences)	Most relevant recent news for outreach context	News API
Website Personalisation Hook	Short text	One sentence that references something specific about the company — used in {Opener} variable	Website AI analysis
Funding Summary	Text	Total raised, last round, date, investors	Crunchbase
ICP Reasoning	Text	Why this company scored the way it did — editable	Internal scoring engine

Custom AI Columns — User-Defined

Users can create any custom AI column by writing a prompt template. The system runs this prompt against each company's available data and writes the result as a new column. This is a freeform intelligence layer.

Example Custom Column	Prompt: "Based on the company description and website, does this company have a dedicated outbound sales team? Answer Yes / No / Unclear and give a one-sentence reason." — Column name: has_outbound_team

Example Custom Column	Prompt: "What is the most relevant pain point this company likely has regarding [client product category]? One sentence." — Column name: likely_pain_point

Custom AI column outputs can be used as filter criteria in Campaign Assembly, as personalisation tokens in email copy (e.g. {{likely_pain_point}}), and as segmentation tags alongside signal tags.

Enrichment Configuration

- Run enrichment automatically on upload — or manually trigger per list
- Select which enrichment tracks to run (scoring, signals, AI columns, or all)
- Configure API sources (Crunchbase key, LinkedIn cookie, BuiltWith key etc.) in workspace settings
- Set re-enrichment schedule — weekly / monthly / on-demand
- Enrichment cost estimator — shows approximate API cost before running
- Partial enrichment — run only on High fit companies to reduce cost

List View — Enriched Output

After enrichment, the list is displayed in a spreadsheet-style view with all columns visible. Users can:

- Sort by any column (ICP Score, signal tag count, funding date, headcount growth)
- Filter by ICP Fit Category, signal tags, employee range, industry, or any custom column
- Save filters as named Segments (each gets a SEGMENT_ID)
- Bulk-edit company records
- Export enriched list as CSV
- Push directly to Campaign Assembly from any saved segment

05 Messaging Studio

AI-generated sequences · variable architecture · A/Z testing · versioning

Purpose

The Messaging Studio is the most product-differentiated module in ConnectaFlow. It produces, for each ICP and persona combination, a complete multi-step email sequence with per-variable A/Z testing architecture — the equivalent of what a senior agency strategist spends two days building manually.

The AI drafts everything. The human edits and approves. The output standard matches the SalesHive playbook format: persona-specific copy, full sequence with 3–5 variants per testable variable, and a clean A/Z test configuration ready for Smartlead.

Email Anatomy — The Canonical Structure

Every email in ConnectaFlow is built from the same named components. This is the agency-defined anatomy that enables precise variable-level testing. Each component is independently editable and testable.

Subject	{Subject}	The email subject line — highest leverage test variable	✅ Primary test axis
Greeting	{Greeting}	Opening salutation — Hi {FirstName}, Hello {FirstName}-	❌ Standardised
Opener	{Opener}	The first sentence — question, statement, or observation. Reader decides to continue or delete based on this.	✅ High leverage
Problem	{Problem}	One specific challenge your buyer is facing — names the pain you are solving	✅ Test which pain resonates
Value Proposition	{ValueProp}	One value statement — what your product does about the problem	✅ Test angles and framing

Story	{Story}	1–3 sentences referencing a specific customer proof point, result, or scenario	✅ Test proof points
Call to Action	{CTA}	The ask — meeting, resource, question. Directness and framing vary significantly.	✅ Test directness vs. softness
Closer	{Closer}	Salutation — Best, Kind regards, Cheers	❌ Standardised
Alias Name	{AliasName}	The sender's name — can be tested (executive vs. SDR)	✅ Test sender persona
Client Company	{Client.Company}	Your company name — fixed	❌ Fixed
Opt-Out	{OptOut}	If you no longer need sales meetings, reply "not interested"	❌ Standardised

A/Z Variable Priority Ladder

Tests are run one variable at a time. This is the discipline the platform enforces — you cannot A/Z test Subject AND Opener in the same campaign. The priority order below reflects agency-validated sequencing: fix higher-impact variables first before testing lower-impact ones.

1	**Subject Line**	3 subject variants — curiosity / pain / direct	Open Rate	3
2	**Opener**	Pain-led vs. insight-led vs. question-based first sentence	Reply Rate	2–3
3	**Problem Framing**	Which specific pain resonates most with this persona	Reply Rate	2
4	**CTA Style**	Direct meeting ask vs. soft question vs. resource share	Positive Reply Rate	3
5	**Value Proposition**	Which product angle lands — cost, speed, quality, risk	Positive Reply Rate	3
6	**Story / Social Proof**	Which customer proof point converts — growth, cost, speed	Positive Reply Rate	3
7	**Sender Alias**	Executive vs. SDR sender persona effect on open + reply	Open Rate + Reply Rate	2

Messaging Set Structure

For each ICP + Persona combination, the Messaging Studio produces one complete Messaging Set with the following layers:

Layer 1 — Persona Card

Auto-generated from GTM Intelligence persona data. Displayed as context inside the Messaging Studio. Not copy — it is the strategic brief the copywriter (or AI) uses.

- Persona name, titles, decision role
- Pain → Solution matrix (3–5 pains, each mapped to product-specific resolution)
- Insight to provide — non-obvious market truth
- Thoughtful opening question — for use in Opener variants
- Common objections + suggested responses

Layer 2 — Sequence Structure

Default 5-step sequence. Configurable to 3 or 7. Agency-validated cadence defaults:

1	**Initial Outreach**	Day 1	First impression + interest	Short — 4 sentences max
2	**Follow-up 1**	Day 4	Reinforce value + try different angle	Short — reference step 1
3	**Follow-up 2**	Day 11	New angle — insight or story	Medium — add proof point
4	**Follow-up 3**	Day 18	Address assumed objection	Short — acknowledge silence
5	**Break-up**	Day 25	Final touch — leave door open	Very short — no pressure CTA

All day timings are configurable. Sequence length is configurable. The defaults reflect agency practice and are editable per campaign.

Layer 3 — Full Copy Per Step

For each sequence step, the system generates the full assembled email PLUS each named component broken out individually for editing and A/Z variant generation.

Assembled Email	Full email with all components populated — preview renders as it would appear in inbox
Subject A/B/C	3 subject line variants — labelled by angle (e.g. "Curiosity", "Pain", "Direct")
Opener A/B/C	2–3 opener variants — labelled by type (e.g. "Pain-led", "Insight-led", "Question")
Problem A/B	2 problem framing variants — different pains from the persona card
Value Prop A/B/C	3 value proposition variants — different product angles
Story A/B/C	3 proof point variants — drawn from client's best customer examples
CTA A/B/C	3 CTA variants — meeting ask / question-based / resource share
Personalisation Tokens	All available tokens listed: {{first_name}}, {{company}}, {{signal_context}}, {{custom_ai_column}}

Layer 4 — Personalisation Tokens

Every AI column generated in the Enrichment Engine is available as a personalisation token in email copy. This enables signal-aware personalisation at scale without manual work.

Example	Subject: "Scaling {company}'s outbound?" | Opener: "I noticed {company} just {signal_context} — typically when companies hit that stage, outbound pipeline becomes the bottleneck." | Where signal_context = "raised a Series B" for funded companies, "added 40 people in 6 months" for growth-stage, etc.

Messaging Studio UI Features

Inline Editing

- Every component editable inline — click any field to edit
- Regenerate any single component without touching the rest
- Tone adjustment slider — more direct ↔ more conversational (re-runs AI on that component)
- Approve / Flag / Regenerate per component

Versioning

- Every saved messaging set gets a MESSAGING_ID with version number

- Full version history — can revert to any previous version
- Label each version (e.g. "Pain-led v1", "Signal-aware v2")
- Compare versions side by side

Approval Workflow

- Mark each sequence step as Draft / In Review / Approved
- Campaign Assembly only allows approved messaging sets to be selected
- Comment threads per component for team review

Persona-Specific Copy

- Each persona gets its own messaging set — VP Sales copy is different from CRO copy
- Signal-specific variants — the system can generate a separate opener for companies with the recently_funded tag vs. standard cold approach
- Play-type adaptation — Competitive Displacement copy is structurally different from Trigger-informed copy — the system knows this

06 Campaign Assembly

Package the experiment · write the hypothesis · push to Smartlead

Purpose

Campaign Assembly is where a named, structured GTM hypothesis is packaged and launched. It connects a list segment to a messaging set and sequence, configures the A/Z test, validates volume, and pushes the complete package to Smartlead via API. Every campaign that leaves ConnectaFlow is a well-defined, traceable experiment.

Campaign Configuration

Campaign Name	Descriptive label — e.g. "ICP01 — VP Sales — Pain-led — Subject Test"
Hypothesis	One sentence required before launch. e.g. "US SaaS VP Sales will reply more to pain-led subject lines than curiosity-based at >3% rate." This is enforced.
ICP	Select linked ICP_ID — determines scoring context for analytics
List Segment	Select from saved segments (filtered by ICP score band + signal tags) — or create new filter inline
Messaging Set	Select approved MESSAGING_ID — persona and ICP must match

Sequence	Select SEQUENCE_ID — step count, timing, sending windows
A/Z Test Variable	Select one variable to test this campaign — system enforces one variable only
A/Z Variants	Select which variants run — minimum 2, maximum 4
Volume Check	System calculates contacts per variant — warns if below 200, blocks if below 50
Sender Configuration	Alias name(s), sending email(s) — pulled from Smartlead workspace
Send Schedule	Start date, daily volume cap, sending hours — passed to Smartlead
Smartlead Campaign Name	Auto-generated or custom — used for Smartlead-side reference

Volume Discipline

Rule	Minimum 200 contacts per variant for statistically meaningful A/Z results. The platform warns below this threshold. Users can override with explicit acknowledgment. The system never silently runs an underpowered test.

The volume check also calculates expected results based on industry benchmarks and prior campaign performance from the same ICP — helping users understand if the segment is large enough to expect statistically significant results.

Smartlead Push

ConnectaFlow pushes to Smartlead via API. The payload includes:

- Contact list with all enrichment columns as merge fields — any column from the enriched list is available as a personalisation token in Smartlead
- Sequence steps with full copy — each step assembled from approved components
- A/Z test configuration — variant labels, split logic, and test variable identifier
- Sending schedule and volume caps
- Campaign metadata — CAMPAIGN_ID, ICP_ID, MESSAGING_ID, SEGMENT_ID passed as Smartlead tags for analytics retrieval

Separation of Concerns	ConnectaFlow passes everything Smartlead needs. Deliverability management, inbox rotation, bounce handling, and send throttling are owned entirely by Smartlead. ConnectaFlow never interferes with the execution layer.

07 Analytics Module

Pull metrics · slice by strategy · campaign review · next brief

Purpose

The Analytics module pulls campaign performance data from Smartlead and enriches it with ConnectaFlow's structural metadata — ICP, signal segment, messaging variant, play type. The result is not just a metrics dashboard but a strategic learning layer: you can see not just your open rate but which ICP, which signal, which messaging angle, and which A/Z variant drove it.

Metrics Retrieved from Smartlead

Emails Sent	Total emails sent across all steps and variants
Emails Delivered	Sent minus bounced — deliverability rate
Unique Opens	Individual contacts who opened at least one email
Open Rate	Unique opens / delivered
Total Replies	All replies regardless of sentiment
Reply Rate	Total replies / delivered
Positive Replies	Replies categorised as interested, requesting more info, or booking
Positive Reply Rate	Positive replies / delivered — primary success metric
Meetings Booked	Calendar invites confirmed — synced from Smartlead booking tracking
Bounce Rate	Hard and soft bounces as % of sent — deliverability health signal
Unsubscribes	Opt-out rate — flagged if above threshold
Step-level performance	Open and reply rates broken down per sequence step
A/Z variant performance	Per-variant metrics for the test variable — winner declared at configurable confidence threshold

Analytics Dimensions — The Strategic Slices

Every metric can be sliced across these dimensions. This is what makes ConnectaFlow analytics different from a standard campaign dashboard.

ICP	Which ICP performs best overall?	Informs whether to invest more list-building effort in this ICP or move to a new one
Signal Segment	Do recently_funded companies reply at a higher rate?	Confirms whether specific signals are worth the enrichment cost to detect
Messaging Angle	Does pain-led or insight-led copy perform better for this persona?	Tells the Messaging Studio which angle to default to in future campaigns
Sequence Step	Where in the sequence do most replies come?	Informs optimal sequence length — are steps 4 and 5 earning their send volume?
A/Z Test Variable	Which subject line / opener / CTA won?	The primary experimental result — winner gets locked into next campaign
Play Type	Which motion type books the most meetings?	Informs which play types to scale and which to retire
Persona	Are we reaching the right title? Do CROs or VPs respond more?	Validates persona selection in GTM Intelligence
Sender Alias	Does sending from a founder name vs. SDR name improve opens?	High leverage test — often 2–3x difference in open rate

A/Z Test Result Declaration

- Default confidence threshold: 95% — configurable per workspace
- Winner declared automatically when threshold is reached
- Losing variants paused in Smartlead via API
- Winning variant tagged as "Confirmed Winner" on the VARIANT_ID
- Winning variant auto-suggested as default in next campaign's messaging set for this variable
- Inconclusive results (insufficient volume) flagged with recommendation to continue or increase segment size

Campaign Review Report

After a campaign reaches sufficient data volume (configurable — default 2 weeks or 500 delivered), the system generates a Campaign Review Report. This is a structured, human-readable summary designed to brief the next campaign.

- Hypothesis — was it confirmed, refuted, or inconclusive?

- Performance vs. benchmark (industry average and workspace average)
- A/Z test result and winning variant
- Top performing signal segment — which companies in the list responded most
- Step-level drop-off analysis
- Recommended next hypothesis — AI-generated, based on results
- Suggested changes to ICP, segment, messaging, or play type for next campaign

08 Workspace & Settings

Multi-client · team access · integrations · configuration

Workspace Structure

Each workspace in ConnectaFlow represents one client or brand running outbound campaigns. All GTM Intelligence, lists, enrichment data, messaging sets, and campaigns are scoped to a workspace.

- Multiple workspaces per account (agency use case — one workspace per client)
- Workspace-level settings — Smartlead connection, API keys, enrichment sources, A/Z confidence thresholds
- Team members with role-based access — Admin / Strategist / Analyst / View-only
- Workspace activity log — all changes tracked with user and timestamp

Integrations

Smartlead	API (bidirectional)	Campaign push, metrics pull, A/Z result sync — primary execution integration
Crunchbase	API (read)	Funding event detection for recently_funded signal
LinkedIn	Unofficial API / cookie	Job posting detection, headcount delta, leadership changes
BuiltWith / Wappalyzer	API (read)	Tech stack detection for tech_stack_match signal
News API / Apollo	API (read)	News mentions, expansion announcements
Clay	Manual (CSV import)	Clay-enriched lists imported as CSV — columns mapped on upload
Apollo	Manual (CSV import)	Apollo-exported lists imported as CSV
Webhook (outbound)	Webhook	Push enrichment outputs or campaign results to external systems

09 MVP Scope

What ships first · what is V2 · build priority

MVP Must-Haves

The MVP must prove one thing: a GTM team can go from zero context to a live, structured outbound campaign — with scored lists, signal-tagged segments, and tested messaging — faster and more intelligently than they could with disconnected tools.

GTM Intelligence — full input layer	✅ MVP	All sections A–E
ICP, persona, trigger, signal generation	✅ MVP	AI-generated, human-editable
Motion intent definition	✅ MVP	Strategic only — no copy
GTM Strategy Brief export	✅ MVP	PDF + shareable link
List Sourcing Guide per ICP	✅ MVP	Copyable Apollo/Clay filter output
CSV upload + column mapping	✅ MVP	Required fields validation
Deduplication	✅ MVP	Domain-based primary key
ICP Scoring (structured + contextual)	✅ MVP	Core of enrichment value
Signal detection — job postings, funding	✅ MVP	2–3 signals at MVP
Built-in AI columns (10 columns)	✅ MVP	Company summary, model, category etc.
Custom AI columns (user-defined)	✅ MVP	Freeform prompt → column — high differentiator
Enriched list view with filters	✅ MVP	Sort, filter, save segments

Messaging Studio — AI sequence generation	✅ MVP	5-step default, persona-specific
Full email anatomy with named components	✅ MVP	All 11 variables
A/Z variant generation per component	✅ MVP	2–3 variants per testable variable
Inline editing + approve/reject	✅ MVP	Human approval layer
Messaging set versioning	✅ MVP	Basic version history
Campaign Assembly — full config	✅ MVP	All fields + hypothesis requirement
Volume check with warning	✅ MVP	Warn < 200, block < 50
Smartlead API push	✅ MVP	Contact list + sequence + A/Z config
Analytics — Smartlead metrics pull	✅ MVP	All core metrics
Analytics — ICP/signal/variant slicing	✅ MVP	Core strategic dimensions
A/Z winner declaration	✅ MVP	95% confidence threshold
Campaign Review Report	✅ MVP	Structured summary
Multi-workspace (agency mode)	✅ MVP	One workspace per client
Play Type Catalogue	✅ MVP	12 built-in play types
Real-time signal-to-send automation	❌ V2	Out of scope by design
CRM integration	❌ V2	Salesforce, HubSpot
LinkedIn execution	❌ V2	Outreach channel extension
Multi-language copy generation	❌ V2	International expansion
Direct Apollo/Clay API sync	❌ V2	CSV import sufficient at MVP
Contact-level enrichment at upload	❌ V2	Company-level first
Custom reporting dashboards	❌ V2	Standard slices sufficient at MVP

10 Build Priority & Engineering Notes

Sequence of development · technical considerations

Recommended Build Sequence

1	**Data Model + ID Architecture**	All downstream features depend on clean ID relationships. Build this first, get it right, never refactor it.
2	**GTM Intelligence Module**	Everything downstream reads from this. No list scoring, messaging, or campaign is meaningful without a well-structured GTM Intelligence foundation.
3	**List Ingestion + Column Mapping**	Core data pipeline. CSV → normalised company records. Domain dedup. ICP tagging.
4	**ICP Scoring Engine**	Structured fit first (deterministic, fast). Contextual fit second (AI-assessed, slower). Both write to company record.
5	**Signal Detection (2–3 signals at MVP)**	Job posting signal + funding signal as minimum viable. Each signal source is a separate integration — build as pluggable pipeline.
6	**AI Column Engine**	Custom column architecture is high-value and high-differentiation. Prompt → per-company result → stored as column. Build the runner first, then add built-in column prompts.
7	**Enriched List View**	Table UI with sort, filter, segment save. This is the UX layer over the enrichment engine — build after the data is reliable.
8	**Messaging Studio**	Most product-differentiated module. AI generation using GTM Intelligence context. Component-level editing. Version control. Approval states.
9	**Campaign Assembly + Smartlead Push**	Packaging layer. Reads from all upstream modules. Smartlead API is well-documented — the complexity is in the payload construction, not the API call.
10	**Analytics Pull + Slicing**	Smartlead metrics pull + join against ConnectaFlow metadata (ICP_ID, SIGNAL_ID, VARIANT_ID). Build the join layer carefully — this is the core of the analytics value.

Key Technical Considerations

AI Generation Quality

The quality of AI-generated ICPs, personas, and email copy is directly proportional to the richness of the Section A input. The system should prompt users to be specific — vague inputs produce generic outputs. Consider a "context quality score" that tells users how complete their GTM Intelligence setup is before they proceed.

Enrichment Pipeline Architecture

Build enrichment as an async queue. Large lists (1,000+ companies) should not block the UI. Show real-time progress per company. Allow partial enrichment — the user should be able to work with the first 100 scored companies while the rest are still processing.

Smartlead Sync

ConnectaFlow passes metadata tags (ICP_ID, SEGMENT_ID, MESSAGING_ID, VARIANT_ID) to Smartlead campaigns and contacts. When pulling analytics, use these tags to join Smartlead data back to ConnectaFlow objects. This join is the engine of the analytics layer — design it deliberately.

A/Z Test Integrity

Enforce single-variable testing at the data model level, not just the UI. A campaign record has one AZ_TEST_ID with one test variable. Multiple variable changes in the same campaign must be blocked at the API layer, not just warned about in the frontend.

Custom AI Columns — Rate Limiting

Custom AI column runs against large lists can be expensive and slow. Implement cost estimation before run, rate limiting per workspace, and incremental processing (run on High fit companies first). Cache company-level results so the same company is not re-enriched unnecessarily across multiple lists.